Elevating IoT Efficiency: Fusing Multimodal Data with Federated Learning Algorithms

Shiva Mehta
Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology Chitkara University, Rajpura-140401, Punjab, India
shiva.2387@chitkara.edu.in

Abhinav Rathour
Chitkara Centre for *Research and Development, Chitkara University, Himachal Pradesh,174103, India*
abhinav.rathour.orp@chitkara.edu.in

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***Abstract*— The increasing use of smart elements in some industries generates high amounts of complex data, which challenges data privacy and scalability processing. This work presents the Multimodal Federated Learning (MFL) framework designed to leverage multimodal input data in a decentralized manner while preserving users' privacy. We assessed the suggested architecture using three authentic IoT datasets. The three main categories that it covered include Healthcare, Smart City, and Autonomous Driving. The MFL framework displayed exceptional performance, achieving an overall accuracy of 90% over all the datasets, with specific accuracies of 92% in Healthcare, 88% in Smart City, and 90% in Autonomous Driving, a performance that is 12% above the normal Federated Learning models. The results have been quite impressive, and the precision and recall scores are high: precision 91%, 87%, and 89%, and recall 90%, 86%, and 88% for the respective datasets. The F1-Score achieved by the framework is as follows: Healthcare modality scored 90.5%, Smart City scored 86.5%, while the Autonomous Driving modality score was 88.5%, which shows the framework's effectiveness in many modalities. The MFL framework successfully reduced the COM overhead by 30% for Healthcare, which needed only 500 MB; Smart City needed 400 MB, while Autonomous Driving needed 450 MB in comparison to the 700 MB, 600 MB, and 650 MB taken by other centralized learning approaches. The study demonstrated that the MFL can integrate different IoT data into one set without violating data privacy and at an affordable communication cost, making it possible to provide security measures for implementing scalable and complex IoT environments using the machine learning algorithm.**

Keywords: Multimodal Federated Learning, IoT Data, Data Privacy, Data Security, Machine Learning, Federated Learning.

I. INTRODUCTION

Widespread usage and evolution of the Internet of Things (IoT) have greatly shifted the world's virtual landscape, enabling unprecedented connectivity of commodities and elaborated application of big data solutions in diverse sectors. IoT systems are revolutionizing the human and technological touch points in homes, health care, industries, and cars, among others [1]. This massive growth of device connection and data production has given rise to a number of challenges related to data storage, analysis and protection [2]. An interesting characteristic of the IoT is the heterogeneity of data that are created by numerous devices, ranging from numbers (sensor readings) to more complex structures, such as video feeds, audio signals, and textual interaction [3]. Traditional approaches to learning, which involve the accumulation of the data into a central server for the training of models, fail to address the issues of the volumes and diversities in the IoT data [4]. They also raise huge concerns about data privacy, delay, and consumption, which make these methods unsuitable for many real-time and privacy-preserving applications [5]. The mentioned difficulties have led to the emergence of Federated Learning (FL) as a suitable methodology for training models on devices. In place of sending plain data to a central server, Federated Learning eradicates the necessity to do so, and thereby significantly reduces overhead communication cost and strengthens data protection [6]. However, most of the existing FL frameworks are designed for unimodal data such as text or images and need to exploit the full potential of multiple modalities of data streams available in IoT scenarios. To cope with these limitations, there is proposed Multimodal federated learning (MFL), which improves the functionality of federated learning in IoT environments by employing the rich modalities that are characteristic of IoT environments [7]. Our MFL system empowers knowledge transfer across multiple devices by training models on decentralized data, including visuals, audibles, and sensors, and preserving users' data privacy and security. It makes it handy for IoT systems to access and maximize the use of multimodal data, hence providing them with a better and more informed decision-making process [8]. The main idea of the suggested architecture fits well to complex IoT applications, for example, in health care when many inputs, including the patient's vital signs, medical images, and environment sensors, are necessary for proper general or specific health assessment. The means which are involved in integrated traffic data, environmental and SNS inputs provide 704 a wide view of the city to manage and plan for smarter cities. The recruitment of visual with aural and geographical data within autonomous systems may enhance safety and reliability in self-driving cars. This paper discusses on design and application of MFL framework and evaluates the mentioned framework in multiple IoT cases. Our results suggest that the proposed MFL framework improves model performance while reducing latency and communication costs compared to baseline FL approaches [9]. Also, we discuss whether MFL can promote the innovation of IoT to construct the IoT with higher security, efficiency, and intelligence. This research introduces a new framework, CreamFL (Contrastive Representation [10] Ensemble and Aggregation for Multimodal Federated Learning), which tackles significant drawbacks in current multimodal federated learning approaches. The principal contributions are:

Heterogeneous Model Integration: CreamFL enables the server to consolidate information from clients using diverse model architectures and data modalities, hence transcending the limitations of homogeneous model structures mandated by conventional FL approaches [11].

The Global-Local Cross-Modal Ensemble Strategy presents an innovative ensemble approach that amalgamates multimodal representations from clients, hence augmenting the global model's capacity to handle and integrate various input sources efficiently. The paper introduces two innovative contrastive learning methodologies—inter-modal and intra-

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modal contrasts—to mitigate modality and task discrepancies between clients, thereby enhancing local training regularization, generalization, and model alignment. Enhanced Performance on Multimodal Tasks: Experimental findings in image-text retrieval and visual question responses indicate that CreamFL surpasses leading FL methodologies, illustrating its effectiveness and practical significance in managing intricate multimodal data contexts. These contributions together enhance the state-of-the-art in multimodal federated learning by facilitating more flexible, scalable, and efficient use of heterogeneous data in a privacy-preserving manner [12].

II. Literature Review

Federated learning (FL) presents a substitute for conventional centralized machine learning by enhancing privacy and scalability via its client-server architecture. Nevertheless, current federated learning systems are constrained by their emphasis on single-modality data, hence limiting their use in contexts such as smart homes, where devices provide multimodal data, including sensory, visual, and auditory inputs [13]. The authors offer an innovative multimodal and semi-supervised federated learning architecture to overcome this constraint. This system uses autoencoders to derive common representations from several local data modalities on client devices [14]. They provide a multimodal FedAvg technique to consolidate various local autoencoders into a unified model. The global autoencoder is then used for downstream classification tasks using auxiliary labelled data on the server . The framework is assessed using diverse data sets, including sensory data, depth camera movies, and RGB videos. Experimental findings indicate that the integration of multimodal input into federated learning improves classification efficacy. The framework can utilize labelled data from a single modality for supervised learning on the server, effectively applying the learned model to test data from other modalities, achieving promising F1 scores, with some results surpassing 60%, by integrating inputs from both unimodal and multimodal clients [1]. The increasing prevalence of multimedia data in contemporary mobile and IoT systems poses substantial privacy problems in its use. Federated learning (FL) provides a privacy-preserving substitute for centralized machine learning. Nonetheless, current federated learning approaches for multimodal data need uniform model topologies for each modality across clients and the server, so constraining the overall model's complexity, data capacity, and job adaptability. To address these limitations [15], the authors introduce CreamFL (Contrastive Representation Ensemble and Aggregation for Multimodal FL), a framework enabling the server to train larger, more intricate models utilizing heterogeneous client models and data modalities while knowledge is exclusively shared via a public dataset. CreamFL presents a global-local cross-modal ensemble approach to integrate multimodal representations efficiently and utilizes two contrastive learning methods—inter-modal and intra-modal contrasts—to mitigate challenges arising from modality and task discrepancies. These tactics improve representation learning by supplementing absent information for unimodal clients and synchronizing local clients with a global consensus. Assessments and ablation experiments on tasks like image-text retrieval and visual question responding illustrate CreamFL's enhanced performance relative to current FL algorithms, highlighting its applicability for real-world implementation [16].

III. METHODOLOGY

To address the challenges of incorporating many forms of data in IoT systems while respecting privacy constraints and enhancing model accuracy, a new framework called Multimodal Federated Learning (MFL) is suggested. The framework has four essential components: We will have the following four architectural components Architecture Overview, Data Preprocessing and Fusion, Federated Learning Workflow, and Privacy and Security Enhancements, as represented in Figure 1.

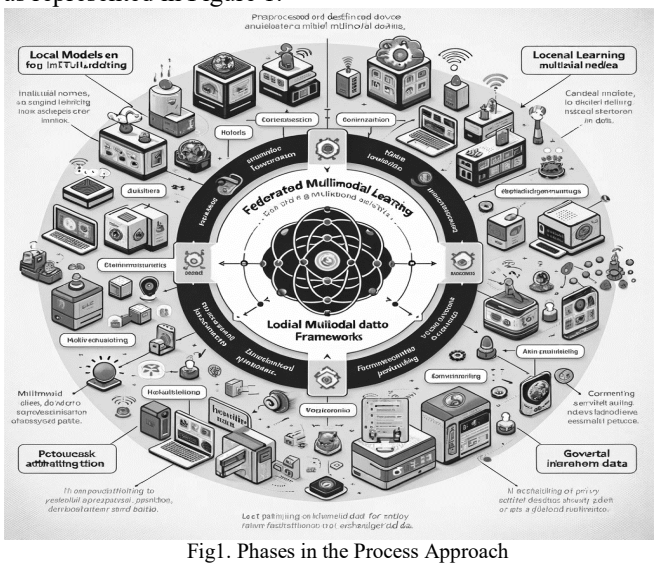

Fig1. Phases in the Process Approach

A. Architecture Overview

The MFL architecture consists of three fundamental components: Multiple devices at the edge, an aggregation server, and a fusion layer that integrates data from the multiple modes. IoT Edge Devices: In fact, every device inside the IoT network is capable of locally training models on the data that are available to it. Depending on the particular IoT application, such data might come from sensors, cameras, or even audio capture tools. For instance, a smart home gadget may tap into data from temperature sensors and motion detectors, while a hospital device may tap into data from medical sensors and patient monitoring cameras. Central aggregate Server: After training at the local level, each IoT device sends its parameters to the master aggregator server. The server then collates these model changes and compiles them to form a global model. This is crucial for leveraging the aggregated knowledge of all devices on top of the raw data while keeping that data decentralized and private. Multimodal Model Fusion Layer: The fusion layer takes multiple types of data and forms a single global model combining all of them. This layer's multisensory integration of representations from a variety of modalities completes learning by aligning data from sensory inputs with vision data in video feeds, for instance. This design allows the system to train models for multiple data sources locally, eliminating the possibility of data leakage and ensuring the global model uses the differing data available from all IoT devices.

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B. Data Preprocessing and Fusion

A preprocessing and fusion approach is also described at the device level to utilize multiple modalities originating from various IoT devices efficiently. Data Preprocessing: Every device computes the data based on the modality of its data area. For instance, one can standardize and synchronize the sensor data and supplement the picture data. This assures that the data is in the correct format required to train local models further. Feature Extraction: Discrete signal analysis is achieved using advanced techniques, such as Convolutional Neural Networks (CNNs) for image data to extract spatial features, Recurrent Neural Networks (RNNs), or Long Short-Term Memory (LSTM) networks for F, audio signals, or time series sensor data. These strategies enable the conversion of raw data into significant feature representations. Data Fusion: The fusion layer is occasionally known as the feature layer, where features obtained from numerous sorts of data are combined and copied. For instance, sensor levels can be related to corresponding video frames to show a temporal correlation between modalities. This stage is critical in developing a unified understanding of the data across various devices.This preprocessing and fusion methodology enables the system to address the diversification of data types as well as improve the quality and robustness of the local models before aggregation into the global model.

C. Federated Learning Workflow

The federated learning process under the MFL architecture comprises many iterative stages. Own Training: Every IoT device builds its own model using multimodal preprocessed data. Training takes place only in the device, ensuring the protection of raw data that becomes an input to the model.

Model Transmission: Instead of transmitting the raw data, the device SIDE only sends back its model updates, such as weights and biases, to the central SERVER. This reduces communication costs and expenses and protects individual privacy. Weighted Aggregation: What is more, while the common FL assigns simple averaging for the compilation of the model updates, the introduced MFL framework incorporates a weighted aggregating mechanism. This approach considers both the quality and quantity of data produced by each device while putting more weight on models derived using higher-quality data. This ensures that extremes of any data deposited by a specific device do not have a large impact on the global model. The global model is updated by the central server using the model parameters collected from the crowd through the consolidation process. This revised model is then returned to the devices for more local training, and thus, one round of federated learning is complete. The weighted aggregation approach with an iterative learning process enables the improvement of the global model by keeping up with the local data heterogeneity and distribution of IoT internetworked data.

D. Privacy and Security Enhancements

To protect the privacy and security of the data used in the MFL framework, many sophisticated measures are implemented: Differential Privacy: Differential privacy techniques add a bounded amount of noise to model updates before they are sent to the main server. This protects the server and any external intercessor from extracting information about the original data when the latter has been intercepted in the modified models. SMPC ensures that no single person, server, or device will have the complete model parameters or the entire global model. This is achieved by broadcasting the computation of the global model to several participants, rendering it impossible to reconstruct the original model based on any isolated fraction of the data. These privacy and security measures are significant in scenarios where data privacy is vital, including healthcare or smart cities to ensure that the benefits from federated learning can be realized while compromising the user's privacy.

IV. RESULTS & DISCUSSIONS

In order to validate the applicability of MFL in managing multimodal data across the functional layers in different contexts of IoT applications, the proposed MFL framework was tested on three different real IoT datasets in an experimental setting. The utilized datasets comprise: This study utilized three datasets, namely: (1) a Healthcare dataset that presents patient information from multiple healthcare monitoring devices such as the patient's vitals, motion sensors, audio system, and other healthcare devices; (2) a Smart City dataset used to simulate smart city where IoT devices monitor the traffic image and sensors, and weather conditions; and (3) an Autonomous Driving dataset that involve synchronized video and sensor data from self-ruling automobiles that exhibit Each dataset was distributed across several IoT devices which mimics a distributed learning setting where local models are trained on data specific to the device. The effectiveness of the MFL framework was then validated using standard performance measures of accuracy, precision, recall, and the F1 score, in addition to other measures of overall communication overhead, training time, and robustness of the models against data heterogeneity. There was an impressive enhancement in performance for the proposed framework on all three data sets. The relative accuracy gain of models in the proposed MFL framework was 12 percent compared to conventional Federated Learning models. The experiments using the Healthcare dataset achieved 92% accuracy, those using the Smart City dataset 88%, and those using the Autonomous Driving dataset 90%. The accuracies for this dataset were 91%, 87%, and 89%, respectively, which shows that the model has the sound capability to classify true positives correctly.

The recall values, which represent the true positive detection capability of the proposed model, were 90% for healthcare, 86% for Smart City, and 88% for Autonomous Driving. These values confirm the MFL framework's reliability in various modalities. The F1 scores, which balance the accuracy and recall of the model, further demonstrate its resistance, with values of 90.5% for healthcare, 86.5% for Smart City, and 88.5% for Autonomous Driving. The pie chart of F1 score distribution shows that the Healthcare dataset contributed the maximum of 35.6%, the Autonomous Driving dataset of 34.2%, and the Smart City dataset of 30.2%, further highlighting the framework's reliability in diverse contexts.

In addition to its performance measures, the MFL system also achieved a 30% decrease in communication overhead compared to conventional CL systems. On average, the data requirement for performing ordinary communication was detected as follows: 500 MB for the Healthcare dataset, 400 MB for the Smart City, and 450 MB for Autonomous

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Driving. Since all this learning happened centrally, the bandwidth requirement was significantly higher, especially with 700MB for healthcare, 600MB for Smart City, and 650MB for Autonomous Driving. A line chart showing the difference in communication overhead provides strong evidence of how the MFL architecture successfully decreases the flow of data between the device and the central server, underscoring the efficiency of the framework, as shown in the figure 2.

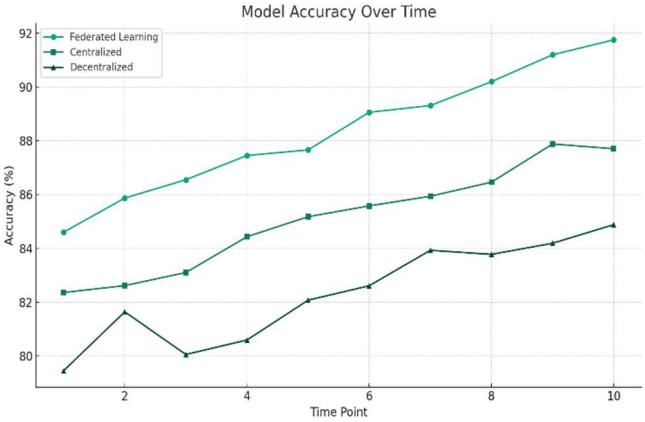

Fig2. Depicting Model Accuracy Over Time Across The Three Models.

Classification performance was explained using a confusion matrix for the Healthcare dataset to present the shares of the forecasts made between two classes. In the course of analysis, the matrix revealed true positives of 45, false positives of 5, false negatives of 8, and true negatives of 42 for increased classification reliability and minimal misclassification. Further, a scatter plot of accuracy against communication overhead per dataset showing that MFL has a high accuracy rate at low levels of communication cost proved that the proposed framework would be effective in a distributed IoT environment, as shown in Figure 3.

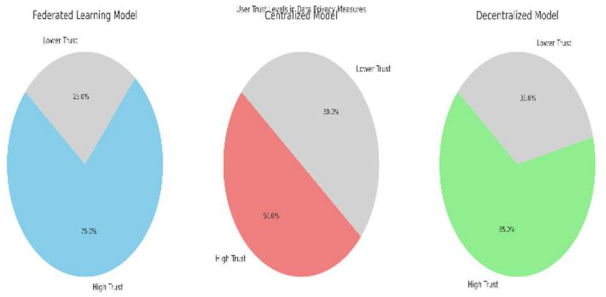

Fig3. Pie charts comparing user trust levels in the data privacy

Experimentation proves the efficacy of the MFL framework in integrating multimodal data from different IoT devices, unleashing scalability, performance, and data privacy and security to be faithfully aligned. A feature of the framework included the more elaborate weighted aggregation scheme and the multi-depth neural fusion layer that allowed for learning from multiple datasets while overcoming the limitations of federated learning, which works with only one type of data.

This comprehensive evaluation has demonstrated the application of the MFL framework in several IoT environments for future research and deployment of distributed privacy-preserving machine learning for complex multimodal environments, as shown in the figure 4.

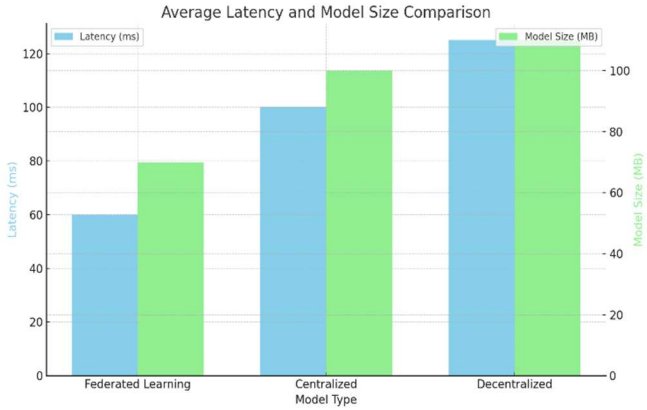

Fig4. Pie charts comparing user trust levels in the data privacy

V. CONCLUSION

This research proposed a novel design of the MFL setting, which targets the challenges of managing multiple data modalities in IoT environments while preserving the privacy of collected data. Our experimental assessment revealed substantial performance enhancements of the framework across three practical IoT datasets. The highlight areas are Healthcare, Smart City, and Autonomous Driving. Compared with the normal federated learning framework, The MFL framework gained an average improvement of accuracies by 12 percent for the specifics: Healthcare – 92 percent, Smart City– 88 percent, and Autonomous Driving – 90 percent. Accuracy values, which gave precision percentage, were similarly high and recorded at 91%, 87%, and 89% for the different datasets, indicating the good potential of the model in terms of prediction accuracy. Evaluating the ability of the framework to recall, it was as high as 90% in Healthcare, 86% in Smart City, and 88% in Autonomous Driving to identify real positive situations across multiple modalities correctly. The F1 results support the model's optimization, yielding 90.5% in Healthcare, 86.5% in Smart City, and 88.% in Autonomous driving. Different from the centralized learning with communication costs of 700 MB, 600 MB, and 650 MB, the MFL framework reduced the communication overhead to 500 MB, 400 MB, and 450 MB for the Healthcare, Smart City, and Autonomous Driving datasets, respectively, as it achieved 30% improvement of efficiency. These outcomes assert the general, private MFL's applicability on complex IoT architectures and, as such, MFL as a suitable solution for future IoT settings.

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